

KEMPTVILLE, ON, Canada

WMO: 713000

Lat: 45.005N

Lon: 75.637W

Elev: 99

StdP: 100.14

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.0	(c) -21.9	(d) -29.2	(e) 0.3	(f) -24.2	(g) -26.5	(h) 0.3	(i) -21.4	(j) 8.5	(k) -5.6	(l) 7.5	(m) -8.1	(n) 2.1	(o) 310	(p) 0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	30.7	22.2	29.1	21.1	27.5	20.4	24.1	28.1	23.0	26.9	21.9	25.7	3.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	17.9	26.0	21.7	16.5	25.3	20.6	15.4	24.2	73.1	28.2	68.6	26.9	64.5	25.8	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.2	6.3	5.5	-28.8	32.8	3.1	1.4	-31.0	33.8	-32.9	34.7	-34.6	35.5	-36.8	36.5
			-29.0	25.6	3.0	1.3	-31.2	26.5	-32.9	27.3	-34.6	28.0	-36.9	29.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-9.1	-7.5	-1.7	6.1	13.0	18.0	20.8	19.9	15.8	8.8	2.1	-4.8
	DBStd	11.62	7.36	6.25	6.19	5.06	4.39	3.72	3.05	3.13	4.19	4.78	5.21	6.50
	HDD10.0	2384	592	489	364	137	19	0	0	0	3	83	239	458
	HDD18.3	4411	851	723	620	367	173	51	10	20	98	297	486	716
	CDD10.0	1247	0	0	2	21	113	241	336	307	178	45	3	0
	CDD18.3	232	0	0	0	1	9	42	87	69	23	1	0	0
	CDH23.3	1983	0	0	1	16	112	367	757	530	189	9	0	2
	CDH26.7	513	0	0	0	2	24	94	219	132	42	0	0	1
Wind	WSAvg	2.4	2.6	2.7	2.8	2.8	2.4	2.1	2.0	1.9	2.0	2.3	2.5	2.5
Precipitation	PrecAvg	902	70	55	62	75	72	87	86	81	87	82	68	76
	PrecMax	1144	128	106	117	152	125	172	184	176	154	175	107	128
	PrecMin	752	30	21	22	12	22	12	15	36	39	18	18	22
	PrecStd	92	27	23	27	35	27	36	38	26	32	34	24	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.4	9.4	18.0	25.5	29.3	31.6	32.9	32.2	30.5	24.5	17.8	13.2
		MCWB	9.5	6.7	12.7	17.5	20.0	23.1	23.2	22.6	22.5	18.9	13.1	10.9
	2%	DB	6.7	5.7	12.8	21.6	26.5	29.4	30.8	29.7	27.6	21.5	15.0	8.4
		MCWB	5.5	3.7	8.7	13.1	17.3	21.7	22.3	21.7	20.8	16.4	12.1	7.1
	5%	DB	3.5	3.4	9.4	17.9	23.9	27.2	29.2	27.9	25.4	18.9	12.4	5.7
		MCWB	2.0	1.7	6.0	11.6	16.1	20.2	21.2	20.6	19.6	15.0	9.7	4.4
	10%	DB	1.2	1.5	6.6	14.9	21.3	25.3	27.5	26.4	23.4	16.2	9.9	3.3
		MCWB	0.2	0.2	3.8	9.6	15.1	18.9	20.6	20.2	18.8	13.0	7.7	2.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	6.9	13.6	18.0	21.5	25.5	26.4	24.9	23.5	20.5	14.5	11.3
		MCDB	10.5	8.0	17.6	22.9	27.7	29.3	30.5	28.7	28.0	23.2	15.8	13.3
	2%	WB	5.4	3.8	9.1	15.1	19.5	23.3	24.3	23.6	22.0	17.5	12.3	7.2
		MCDB	6.4	5.5	11.4	19.4	24.0	27.3	28.4	27.4	26.1	20.4	14.8	8.3
	5%	WB	2.3	1.8	6.5	12.5	17.8	21.7	23.0	22.4	20.7	15.4	10.1	4.4
		MCDB	3.3	3.1	9.0	16.5	21.8	25.7	26.8	26.1	24.3	18.3	11.9	5.6
	10%	WB	0.2	0.4	4.1	10.3	16.1	20.2	21.8	21.3	19.5	13.5	7.8	2.1
		MCDB	1.2	1.5	6.2	14.2	19.5	23.6	25.6	24.7	22.6	15.8	9.6	3.0
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.8	9.6	10.8	11.5	10.9	11.2	11.0	11.4	9.3	8.4	7.7
		MCDBR	10.3	10.4	12.5	16.2	15.3	13.4	13.0	12.8	13.2	12.6	11.9	9.8
		MCWBR	9.7	9.1	8.8	9.6	8.1	6.9	6.2	6.2	7.2	8.1	9.0	8.9
	5% WB	MCDBR	9.9	9.9	11.7	13.5	12.6	11.3	10.9	10.7	11.8	11.3	11.4	9.7
		MCWBR	9.6	9.0	9.1	9.4	8.0	6.7	6.5	6.3	7.1	8.2	9.2	9.1
Clear-Sky Solar Irradiance	taub		0.274	0.291	0.321	0.347	0.380	0.414	0.422	0.406	0.367	0.341	0.308	0.280
	taud		2.395	2.350	2.400	2.406	2.352	2.267	2.248	2.297	2.399	2.469	2.499	2.430
	Ebn at Noon		849	897	914	915	889	856	843	844	852	829	804	802
	Edh at Noon		73	92	102	110	121	132	133	122	102	83	67	64
All-Sky Solar Radiation	RadAvg		1.38	2.31	3.63	4.44	5.22	5.64	5.75	5.07	4.00	2.36	1.46	1.02
	RadStd		0.15	0.23	0.30	0.49	0.53	0.44	0.48	0.23	0.34	0.22	0.16	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page